

ECN 594: Identification and Instrumental Variables

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Plan for today

1. **Recap: Logit, Berry inversion, and identification**
2. Direction of bias
3. Uber example
4. Instrumental variables
5. Elasticities and limitations
6. The demographic interaction model
7. Why demographics help

Recap: Logit, Berry inversion, and identification

- Mean utility: $\delta_{jt} = x_{jt}\beta + \alpha p_{jt} + \xi_{jt}$ ($\alpha < 0$)

- **Berry inversion:**

$$\ln(s_{jt}) - \ln(s_{0t}) = x_{jt}\beta + \alpha p_{jt} + \xi_{jt}$$

- **The identification problem:**

- Firms observe ξ_{jt} when setting prices
- $\Rightarrow \text{Cov}(p_{jt}, \xi_{jt}) > 0 \Rightarrow \text{OLS is biased}$
- Need supply shifters (IVs) to identify demand
- **Today:** Direction of bias, IV strategies, and limitations

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Direction of bias

- If high-quality products have high prices...
- OLS sees: high price, but demand still high (because of ξ)
- OLS concludes: price doesn't hurt demand much
- **Result:** $\hat{\alpha}$ biased toward zero (less negative than truth)

Worked example: Bias direction

- **Question:** You estimate a logit demand model using OLS and get $\hat{\alpha} = -0.3$. A colleague says the true α is likely -0.5 .
- Is this consistent with endogeneity bias? Why?

Worked example: Bias direction

- **Question:** You estimate a logit demand model using OLS and get $\hat{\alpha} = -0.3$. A colleague says the true α is likely -0.5 .
- Is this consistent with endogeneity bias? Why?
- **Answer:** Yes!
 - OLS overstates how much consumers like expensive products
 - So OLS finds a smaller (less negative) price coefficient
 - $-0.3 > -0.5$, so this is exactly what we'd expect

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Case study: Uber surge pricing

- Uber's surge pricing: prices rise when demand is high
- **Why is surge pricing endogenous?**

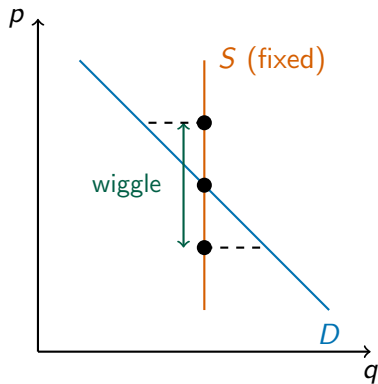
Case study: Uber surge pricing

- Uber's surge pricing: prices rise when demand is high
- **Why is surge pricing endogenous?**
- Surge responds to BOTH:
 - **Supply shocks:** Fewer drivers available → surge up
 - **Demand shocks:** Concert ends, rain starts → surge up
- Can't just regress rides on surge multiplier!
- Same problem as before: observing equilibrium prices

The solution: Uber's price “wiggle” experiment

- Uber's insight: **randomize** prices!
- **Cohen et al. (2016)**: Uber experimentally varied surge multipliers
- Same time, same location → different riders see different prices
- Key: Some riders randomly see 1.9x, others see 2.1x
- Demand conditions are identical, only price differs

Tracing out the demand curve



- Randomization holds supply fixed
- Price “wiggles” up/down randomly
- Traces out points ON the demand curve
- This identifies demand!

Uber experiment: Results

- Result: demand elasticity ≈ -0.5
- **Inelastic!** A 10% price increase $\rightarrow$ only 5% fewer rides
- Why so inelastic?
 - When you need a ride, you *need* a ride
 - Limited substitutes late at night
- **Key lesson:** Experiments give clean identification
- But most industries can't randomize prices...

Why experiments are powerful

- No confounding: price variation is independent of demand shocks
- Clean identification of the demand curve
- **But:**
 - Tech companies can run experiments
 - Traditional industries can't randomize prices
 - Most IO settings require **instrumental variables**

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Instrumental variables: the solution

- Need variables z that are:
 1. **Relevant:** Correlated with price ($\text{Cov}(z, p) \neq 0$)
 2. **Exogenous:** Uncorrelated with ξ ($\text{Cov}(z, \xi) = 0$)
- These are cost shifters or other supply-side variables

Common IVs in demand estimation

Three main approaches used in the literature:

1. **Cost shifters:** Input prices, exchange rates
2. **Hausman IVs:** Prices in other markets
3. **BLP IVs:** Characteristics of competing products

Let's examine each in detail...

IV #1: Cost shifters

- **Idea:** Use variables that shift marginal costs
 - Input prices (steel, labor, energy)
 - Exchange rates (for imported goods)
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 - Steel prices don't make you like/dislike a car
- **Problem:** Cost data rarely observed!
 - Firms don't disclose detailed cost information
 - Even aggregate input prices may be hard to match to products

IV #2: Hausman instruments

- **Idea:** Use prices of the same product in *other markets*
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- **Idea:** Use prices of the same product in *other markets*
 - Coca-Cola price in Chicago as IV for Coca-Cola price in Phoenix
- **Why it works:**
 - Same product has similar costs across markets
 - Cost shocks (ingredients, shipping) affect prices everywhere
- **Key assumption:** Demand shocks ζ_{jt} are independent across markets
 - Chicago's unobserved demand factors don't affect Phoenix

Worked example: When do Hausman IVs fail?

- **Question:** You want to use Pepsi prices in Denver as an IV for Pepsi prices in Phoenix. When might this fail?

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- **Answer:**
- Hausman IVs assume ξ_{jt} is independent across markets
- **This fails when:**
 - National advertising campaigns affect all markets
 - Nationwide product reformulation ("New Coke")
 - Coordinated promotions or seasonal demand patterns
- **Relevance is OK:** Cost shocks do affect prices everywhere
- **Exogeneity is the problem:** Demand shocks may also be correlated!

IV #3: BLP instruments

- **Idea:** Use characteristics of *other products* in the market
- **Key assumption:**

$$E(\xi_{jt}|x_t) = 0$$

“Observed characteristics are mean independent of unobserved characteristics”

- **Common BLP instruments:**
 1. Own product characteristics (not price!)
 2. Sum of characteristics of *own-firm* products
 3. Sum of characteristics of *competitor* products
- **Intuition:** More/closer competitors $\rightarrow$ more competition $\rightarrow$ lower prices

BLP instruments: Example

- Consider estimating demand for the Honda Civic
- **BLP instruments could include:**
 - Civic's own characteristics: horsepower, MPG, size
 - Sum of Honda's other cars' characteristics (Accord, CR-V)
 - Sum of competitors' characteristics (Corolla, Mazda3, etc.)

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 - Sum of Honda's other cars' characteristics (Accord, CR-V)
 - Sum of competitors' characteristics (Corolla, Mazda3, etc.)
- **Why exogenous?**
 - Toyota's decision about Corolla's horsepower doesn't affect Civic's unobserved quality ξ
- **Why relevant?**
 - If Toyota improves Corolla, Honda faces more competition
 - This pressures Honda to lower Civic's price

Worked example: Evaluating an IV

- **Question:** A researcher proposes using gasoline prices as an IV for car prices. Is this valid?

Worked example: Evaluating an IV

- **Question:** A researcher proposes using gasoline prices as an IV for car prices. Is this valid?
- **Relevance:** Higher gas prices $\rightarrow$ higher operating costs $\rightarrow$ might affect car prices? Maybe weakly.
- **Exogeneity:** Do gas prices affect car quality ζ_{jt} ?
 - Gas prices affect *demand* for fuel-efficient cars
 - This might shift which cars look “good” to consumers
 - Potentially problematic!
- **Verdict:** Probably not a great IV

Summary: IV conditions

- For z to be a valid IV:
 1. **Relevant:** z must predict prices
 - Can test this! Run first-stage regression
 2. **Exogenous:** z must not affect demand directly
 - Cannot test this directly (requires economic reasoning)
- This is the standard IV framework from econometrics
- Applied to demand estimation: use supply-side variation

What makes a good IV? Summary

Instrument	Relevance	Exogeneity
Cost shifters (input prices)	Strong	Usually OK
Hausman IVs (prices elsewhere)	Strong	Debatable
BLP IVs (competitor chars)	Medium	Usually OK
Experiments (Uber wiggle)	Perfect	Perfect

- No perfect instrument in observational data
- **Best practice:** Use multiple IVs, check robustness
- **Experiments** are gold standard but often infeasible

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From estimation to elasticities

- We've seen how to estimate demand with logit
- **Why do we care?**
- Demand estimates let us compute **elasticities**
 - How sensitive are consumers to price changes?
 - How do consumers substitute between products?
- These are crucial for policy: mergers, pricing, welfare
- **But:** the logit model has important limitations...

Logit elasticities

- Given shares $s_{jt} = \frac{\exp(\delta_{jt})}{1 + \sum_k \exp(\delta_{kt})}$
- We can derive price elasticities:

$$\eta_{jj} = \frac{\partial s_{jt}}{\partial p_{jt}} \frac{p_{jt}}{s_{jt}} = \alpha p_{jt} (1 - s_{jt}) \quad (\text{own-price})$$

$$\eta_{jk} = \frac{\partial s_{jt}}{\partial p_{kt}} \frac{p_{kt}}{s_{jt}} = -\alpha p_{kt} s_{kt} \quad (\text{cross-price})$$

- Note: $\alpha < 0$, so own-price elasticity is **negative** (as expected)

Limitation #1: Own-price elasticity

- Own-price elasticity: $\eta_{jj} = \alpha p_{jt}(1 - s_{jt})$
- With many products: $(1 - s_{jt}) \approx 1$
- So: $\eta_{jj} \approx \alpha p_{jt}$
- **Implication:** Own-price elasticity is proportional to price!

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- So: $\eta_{jj} \approx \alpha p_{jt}$
- **Implication:** Own-price elasticity is proportional to price!
- Lower-priced products have more *inelastic* demand
- Higher-priced products have more *elastic* demand
- **Question:** Is this realistic?
 - Generic cereal (low price) should be more elastic than organic cereal?
 - Budget airline (low price) more inelastic than first class?

Limitation #2: Unrealistic substitution patterns

- Cross-price elasticity: $\eta_{jk} = -\alpha p_{kt} s_{kt}$
- This doesn't depend on product j at all!
- **Implication:** All products have the *same* cross-elasticity with product k

Limitation #2: Unrealistic substitution patterns

- Cross-price elasticity: $\eta_{jk} = -\alpha p_{kt} s_{kt}$
- This doesn't depend on product j at all!
- **Implication:** All products have the *same* cross-elasticity with product k
- Example: BMW raises its price
 - Logit says: same fraction go to Mercedes as to Honda Civic!
 - Reality: BMW consumers mostly switch to Mercedes, Audi
- Substitution depends on **market share**, not product similarity

Diversion ratios in logit

- When product k 's price rises, where do consumers go?
- **Diversion ratio** from k to j :

$$D_{k \rightarrow j} = \frac{\partial s_{jt} / \partial p_{kt}}{\partial s_{kt} / \partial p_{kt}} = \frac{s_{jt}}{1 - s_{kt}}$$

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Diversion ratios in logit

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- **Why this matters:**
 - Merger analysis: which products are close substitutes?
 - Product positioning: who are my real competitors?
 - Logit can't answer these questions well

Why logit has unrealistic substitution

- The problem comes from the **logit error structure**
- Key assumption: ε_{ijt} are *independent* across products
- This means: your taste shock for BMW is unrelated to your taste shock for Mercedes
- The model doesn't "know" that BMW and Mercedes are similar

Why logit has unrealistic substitution

- The problem comes from the **logit error structure**
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- This means: your taste shock for BMW is unrelated to your taste shock for Mercedes
- The model doesn't "know" that BMW and Mercedes are similar
- **Solution:** Allow for consumer heterogeneity
 - High-income consumers prefer both BMW and Mercedes
 - When BMW's price rises, these consumers switch to Mercedes

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The demographic interaction model

- We estimate a version of **Berry, Levinsohn & Pakes (1995)** — “BLP”
- Allow preferences to vary across consumers:

$$u_{ijt} = x_{jt}\beta_i + \alpha_i p_{jt} + \zeta_{jt} + \varepsilon_{ijt}$$

- β_i, α_i : consumer i 's taste for characteristics and price
- **Key question:** What drives heterogeneity in β_i and α_i ?
- **Our approach:** Observed demographics

$$\beta_{ik} = \beta_k + \sum_{\ell=1}^L \pi_{k\ell} D_{i\ell}, \quad \alpha_i = \alpha + \sum_{\ell=1}^L \pi_{\alpha\ell} D_{i\ell}$$

- $D_{i\ell}$: demographic ℓ for consumer i (income, age, family size, etc.)

Rewriting in terms of mean utility

- Substituting the expressions for β_{ik} and α_i :

$$u_{ijt} = \underbrace{x_{jt}\beta + \alpha p_{jt} + \zeta_{jt}}_{\delta_{jt} \text{ (mean utility)}} + \underbrace{\sum_{k,\ell} \pi_{k\ell} D_{i\ell} x_{jt,k}}_{\mu_{ijt}} + \varepsilon_{ijt}$$

- **Mean utility** δ_{jt} : part common to all consumers
 - Contains “**linear parameters**” (β, α) — enter linearly in δ
- **Individual deviation** μ_{ijt} : how consumer i differs from average
 - Contains “**nonlinear parameters**” ($\pi_{k\ell}$) — enter nonlinearly via shares
- Note: $\alpha < 0$ (higher price $\rightarrow$ lower utility)

From utility to demand

- **Individual choice probability** (with T1EV errors):

$$s_{ijt} = \frac{\exp(\delta_{jt} + \mu_{ijt})}{1 + \sum_{k=1}^J \exp(\delta_{kt} + \mu_{ikt})}$$

- **Market share** (aggregate over consumers):

$$s_{jt} = \frac{1}{I} \sum_{i=1}^I s_{ijt}$$

- This is **demand**: the fraction of consumers who buy product j
- Key: shares depend on mean utilities δ and demographics D_i

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Elasticities with demographic interactions

- **Elasticities with heterogeneous consumers:**

$$\eta_{jkt} = \begin{cases} \frac{p_{jt}}{s_{jt}} \sum_i \alpha_i s_{ijt} (1 - s_{ijt}) & \text{if } j = k \\ -\frac{p_{kt}}{s_{jt}} \sum_i \alpha_i s_{ijt} s_{ikt} & \text{otherwise} \end{cases}$$

- s_{ijt} : probability that consumer i purchases product j
- **Key insight:** Cross-elasticity now depends on $\sum_i s_{ijt} s_{ikt}$
 - Products that attract *similar consumers* have higher cross-elasticity
 - BMW and Mercedes attract similar (high-income) consumers $\Rightarrow$ closer substitutes
- Correlation in μ_{ijt} and μ_{ikt} drives realistic substitution patterns

Classic example: Nevo (2001) - Cereal demand

- “Measuring Market Power in the Ready-to-Eat Cereal Industry”
- Estimated demand for cereal using demographic interactions
- Key findings:
 - High-income households less price-sensitive
 - Families with children prefer kid-targeted cereals
 - Without demographics: wrong substitution patterns
- Result: Basic logit would underestimate markups!



Limitations: Demographics vs Mixed Logit

- **Demographics model** (what we use):
 - Heterogeneity only from observed characteristics
 - Substitution still restrictive within demographic groups
 - Simple and tractable
- **Mixed logit / random coefficients** (advanced):
 - Heterogeneity from unobserved factors too
 - Fully flexible substitution patterns
 - Computationally intensive (requires simulation)
- For this course: demographics are sufficient
- Know that mixed logit exists for research

Key Points

1. **Price is endogenous:** Firms observe ξ_{jt} when pricing $\rightarrow$ OLS biased toward zero
2. **Solution:** Instrumental variables (cost shifters, BLP IVs, Hausman IVs)
3. **Logit limitation:** Cross-elasticity doesn't depend on $j \Rightarrow$ unrealistic substitution
4. **Demographic model:** $u_{ijt} = \delta_{jt} + \sum_{k,\ell} \pi_{k\ell} D_{i\ell} x_{jt,k} + \varepsilon_{ijt}$
5. Demographics $\Rightarrow$ heterogeneous preferences $\Rightarrow$ realistic substitution